



भारतीय
प्रौद्योगिकी
संस्थान
काशी हिन्दू विश्वविद्यालय



INDIAN
INSTITUTE OF
TECHNOLOGY
BANARAS HINDU UNIVERSITY

I-DAPT-HUB FOUNDATION @ IIT (BHU) Varanasi

Call for Innovative and Entrepreneurial Project Proposals Under

Technology Incubation Hub (TIH) of National Mission on Interdisciplinary Cyber Physical Systems (NMICPS) for Data Analytics and Predictive Technologies

Preamble

Interdisciplinary Data Analytics and Predictive Technology (I-DAPT) at IIT (BHU) under Technology Incubation Hub (TIH) targets to leverage the Research and Engineering capabilities of Sustainable Development Goal 7 and achieve the mandate of the National Mission on Interdisciplinary Cyber Physical Systems. The I-DAPT at IIT (BHU) promotes the progression of Technology Readiness Level (TRL) for the development of indigenous systems mainly in the domain areas, i.e. Telecommunication, Power, Defense Research and Development, Road Transport and Highways, Health and Family Welfare including other areas like Energy and Environment, Agri-Tech etc. The I-DAPT at IIT (BHU) also encourages leveraging other related areas for technology development, innovation, professional education, entrepreneurship, brand building, technology commercialization and product management for disseminating and deploying intellectual property; and for public outreach. At IIT (BHU), I-DAPT under TIH acts as a Section 8 Company to spearhead the activities in "Data Analytics and Predictive Technologies" and other related areas under TIH.

Proposal Call

I-DAPT-HUB FOUNDATION at IIT (BHU) under Technology Incubation Hub (TIH) of National Mission on Interdisciplinary Cyber Physical Systems (NMICPS) is inviting project proposals for a multidisciplinary program on **Data Analytics & Predictive Technology (DAPT)**. A critical aspect is that the DAPT project proposals are at a high level of TRL (**above TLR readiness levels 4**) for (i) concrete development of technologies, products, IPRs, as well as research bases and (ii) Entrepreneurship Development, including Startups and Spin-off companies, along with significant employment generation, techno-commercial incubation and innovation acceleration. The proposals are invited under the following sub-themes of **five areas (A to E) and any other areas (F) that are well-fitted in the NMICPS – IDAPT theme**.

A. Telecommunications

B. Power

- C. Defence Research and Development**
- D. Road Transport and Highways**
- E. Health and Family Welfare**
- F. Other areas (Agritech, Energy and Environment, etc.)**

Some indicative/directed areas are given below; however, the investigators may also choose any other related areas.

A. Telecommunications: The proposal should be in the following (or related) sub-themes, leveraging Design, Applications & Use Cases for DAPT in the next generation telecommunication paradigm:

- i. Industry 4.0 – 5G Hyper-connection Air Interfaces
- ii. Time Sensitive Networking (TSN)
- iii. Hyperconverged - Multi-access Edge Computing (MEC)
- iv. Software Defined Networks (SDN) and Network Function Virtualization (NFV)
- v. Ultra-Reliable Low-Latency Communication (URLLC)
- vi. Deterministic Networks for Critical Infrastructure Supervisory Control
- vii. Massive Machine Type Communications (mMTC): M2M/ V2V/ V2I Communication
- viii. Low Power Wide Area Technologies (LPWA), including NB-IoT enhancements and alike
- ix. Smart City Command and Control Centre – Optimal Design and Performance Test
- x. Intelligent Sensors and Systems
- xi. Any other relevant area

B. Power: The proposal should be in the following (or related) sub-themes:

- i. Mix Energy Source Renewable Integration in Sustainable Smart Power Grid.
- ii. Solar PV Integrated DAPT Driven IoT-Based E-Vehicle Charging Infrastructure.
- iii. Data Analytics and Predictive Technology based Multidirectional Power Management for Smart Homes.
- iv. Environmental Prediction Model for IoT Based Smart Agriculture.
- v. Optimization and Power Management of Energy Storage System.
- vi. Wireless Power Transfer and Energy Harvesting.
- vii. High Power Density High Efficiency Hybrid Processor Technology.
- viii. Smart Power Grid Infrastructure with Cyber Security.
- ix. Power Quality in Standalone and Grid Tied System.
- x. DAPT-based IoT-enabled Demand Response Management in Distribution System.
- xi. Power Optimization in Smart Microgrid Infrastructure.
- xii. Reconfigurable Power Processors for Variable Futuristic Loads.
- xiii. Any other relevant area

C. Defence Research and Development: The proposal should be in the following (or related) sub-themes:

- i. Role of drones and radars for border and underwater surveillance.
- ii. Microwave techniques for imaging, including object detecting radars
- iii. Stealth technique based on advanced polymer composites
- iv. Battery for remote areas in defence.
- v. Explosive detection
- vi. Smart sensors for soldiers (Air, Water and health)
- vii. Biosensors to know the readiness of soldiers
- viii. 5G for defence communications
- ix. Advanced Polymer Composites for Defence Applications.
- x. Data analysis and predictive technology related to defence.
- xi. Supply chain (medicine, food, goods, etc.) in defence.
- xii. Any other relevant area

D. Road Transport and Highways: The proposal should be in the following (or related) sub-themes:

- i. Development of pavement monitoring and management system
- ii. Development of low-cost crash prediction and warning system
- iii. Development of a prototype connected vehicle testbed
- iv. Development of pedestrian navigation and mapping system
- v. On-road pavement distress monitoring and prediction system
- vi. Low-cost sensors for on-road vehicular emissions monitoring
- vii. Travel behaviour and demand prediction during a pandemic
- viii. Development of regional parking demand prediction and management system
- ix. Models/algorithms/strategies to reduce air pollution and carbon footprint using Intelligent Transportation Systems
- x. Dynamic road user charges/toll and their economic impact
- xi. Any other relevant area

E. Health and Family Welfare: Proposal should be in the following (or related) sub-themes:

- i. Forecasting and Prognostics
- ii. Disease Prevention
- iii. Rehabilitation
- iv. Screening or Diagnosis
- v. Therapy or Treatment
- vi. Family Welfare

- vii. Brain/Mind Health.
- viii. Any other relevant area

F. Other areas – Energy and Environment, Agri-Tech. with DAPT component

Eligibility:

Applicants [Principal Investigator (PI) and Co-Principal Investigator(s)] should be Indian citizens. Foreign nationals (including OCI and NRI) are also eligible to apply, provided they fulfil the eligibility criteria notified by SERB (SERB Funding Guidelines for foreign nationals dated 27th Oct 2016).

The applicant(s) must hold a regular academic/research position in a recognised academic institution, national laboratory, or other recognised R&D institution in India. Proposals are invited under three categories as given below.

Project Category: Category A /Category B / Category C

Category A (TRL3 and above may apply in this category, to achieve TRL 5 and above)	18 months with a maximum budget of 15 Lakhs The project may be considered for extension with additional funds. Concrete plan to achieve a minimum TRL level 5.
Category B (TRL 4 and above may apply in this category, to achieve TRL 6 and above)	Two years with a budget of up to 25 Lakhs and a concrete plan to achieve a minimum TRL level 6. Extendable to the third year based on progress. An industry partnership of 10% of the proposed budget is mandatory. One PI/Co-PI from IIT (BHU) is desirable
Category C (TRL 5 and above may apply in this category, to achieve TRL 7 and above)	Two years with a budget of over 25 Lakhs and a concrete plan to achieve a minimum TRL level 7. Extendable to the third year based on progress. An industry partnership of 20% of the proposed budget is mandatory for up to 50 lakh projects. For a proposed budget of more than 50 Lakhs, an industry partnership of a minimum of 30% of the total proposed budget is mandatory. One PI/Co-PI from IIT (BHU) is desirable.

Human resources requirements in the project:

Manpower in the project may be provided in the form of Comprehensive and Holistic Advancement of National Knowledge Yield and Analytics (CHANAKYA) fellowships. Two types of CHANAKYA fellowships are available: CHANAKYA regular fellowships and CHANAKYA top-up fellowships.

CHANAKYA regular fellowship – In this category, you can appoint the project staff as CHANAKYA fellows according to the rules and regulations of the PI's Institute. This fellowship is only for CATEGORY B and C projects. PI can select the fellows per their institute's rules and regulations and inform I-DAPT for approval/funding.

Other details of the fellowships are as follows.

Nature of Fellowship	Qualification of the candidate	Fellowship	Duration	Remark
Chanakya UG Fellowship - I	Registered UG students	Rs. 4,000/-	Initially six months, and extendable up to two years	Please see the details in the CHANKYA section on the I-DAPT website
Chanakya PG Fellowships	B.Tech./B.E./M.Sc. or equivalent in the relevant areas Preference should be given to GATE/NET, or any other national-level test-qualified students.	INR 12,800/-	Initially six months, and extendable up to two years	The candidate can register in the regular programs (M.Tech./ME) of the PI institute.
Chanakya Doctoral Fellowship	M.Tech./ M.E./ M.Sc. or equivalent in the relevant areas / proposed area of the project Preference should be given to GATE-qualified candidates. A NET or equivalent qualification is mandatory for students with a science background.	JRF - INR 37,000/- + HRA @ 18%, SRF – INR 42,000/- + HRA @ 18%	Initially six months, and extendable up to three years	The candidate can register in the regular program (Ph.D.) of the PI institute.
Chanakya Post-Doctoral Fellowship	Ph.D. in relevant areas / proposed area of the project	INR 80,000/- (Consolidated)	Initially six months, and extendable up to two years	Applicable for CATEGORY C projects only.

CHANAKYA Top-up fellowships – These fellowships are primarily available for regular IIT (BHU) students. However, the I-DAPT can consider requests for Top-up fellowships for students of other institutions if a high TRL (Above 7), technology development, transfer, or potential startup is

visible/expected from the project. The students can avail the Chanakya top-up fellowship along with a regular fellowship of the institute for doing work related to the project.

Fellowship Type	Top-up per month	Remark
Post Doc Top-up	Rs. 15,000/-	Initially for one year, and extendable on the performance basis
Doctoral Top-up	Rs. 10,000/-	Initially, one year and extendable on the performance basis. The Ph.D. students who have completed SOTA are eligible for the top-up fellowship.
PG Top-up	Rs. 5,000/-	One year only. II year M.Tech. Students are eligible for a top-up.
IDD M.Tech. Top-up (from 4th year onwards)	Rs. 5,000/-	One year only. Extendable on the performance basis. 4th year onwards

The UG students (III year onwards and preferably during the UG project) can also be engaged in the project-related work. CHANAKYA UG fellowship of Rs. 4000/- per month may be given to them for an initial period of 6 months and then extended further on a performance basis. A maximum of three UG students may be engaged in the project.

The project's deliverables are expected from the 1st year onwards. The PI has to update the progress of the project on a quarterly basis. The investigators may use the IIT (BHU) Varanasi facility per the institute's terms and conditions.

The project proposal can be submitted at any time, in the format provided on the website idapthub.org. Projects completed in all respects and as per the format will be reviewed in the subsequent meetings for the project evaluation.

The PI or Chanakya Fellows (through PI as mentor) can approach I-DAPT for registration of a startup based on the expected outcome of the project within six months after sanction of the project. The I-DAPT will help/facilitate the PI/ Chanakya Fellows in the Startups registration process. The Startups may apply for more grants through the Startup funding mode of I-DAPT. The I-DAPT will keep a minimum 5% equity in such startups.

In all Patents filed based on the outcome of the project funded by I-DAPT, I-DAPT will be the main or Co-Applicant. PIs may approach I-DAPT for help in patent filing. All technologies developed, sold, and transferred based on I-DAPT funding projects, I-DAPT will be 50% beneficiary or as per prior mutual agreement signed with PI/Co-PI and I-DAPT.

- For any queries, contact manager2.idapt@itbhu.ac.in and cc to projectdirector.idapt@iitbhu.ac.in. Along with online submission, mail a copy of the project to projectdirector.idapt@iitbhu.ac.in.

Important Note: Please submit PART-I, PART-II and PART-III as a single merged file. The signature of the PI and Co-PI is not required in PART II.

Project Format: PART - I

**Call for Innovative and Entrepreneurial Project Proposals
Under
Technology Incubation Hub (TIH) of National Mission on Interdisciplinary Cyber
Physical Systems (NMICPS) for
Data Analytics and Predictive Technologies (DAPT)**

Project Ref No. _____
(to be given by the office)

Details of Investigators

1. Principal Investigator (PI)

Name	Dr. Sangeetha Prasanna Ram
Designation	Associate Professor
Department	Automation and Robotics
Institution	Vivekanand Education Society's Institute of Technology
Postal Address	Hashu Advani Memorial Complex, Collector Colony, Chembur, Mumbai - 400074
E-mail	sangeeta.prasannaram@ves.ac.in
Date of Birth :	13 th August 1976

1.1 Co-Principal Investigator (Co-PIs)

Name	Ms. Venkata Ramya Tirumalasetti
Designation	Assistant Professor
Department	Automation and Robotics
Institution	Vivekanand Education Society's Institute of Technology
Postal Address	Hashu Advani Memorial Complex, Collector Colony, Chembur, Mumbai - 400074
E-mail	ramya.t@ves.ac.in
Date of Birth :	9th June 1987

1.2 Co-PI / Joint PI from IIT (BHU) [Mandatory for Category B and C projects]

Name	--
Designation	
Department	
Institution	
Postal Address	
E-mail	
Date of Birth :	

2. CVs of PI and Co-PIs (Brief CV includes Google Scholar / IIRIS / Institute website profile)

CV of PI (Google Scholar): <https://scholar.google.co.in/citations?hl=en&user=jL7c8ZoAAAAJ>

CV of PI (Institute website profile): <https://vesit.ves.ac.in/faculty/INST>

CV of Co-PI (Institute website profile): <https://vesit.ves.ac.in/faculty/INST>

3. Ten key publications (Full Reference with Impact Factor) and Patents (IPR) of the Investigators pertaining to the theme of the proposal during the last 5 years

- **Key Publications:**

1. **Sangeetha Prasanna Ram**, Jayalekshmi Nair, Suryanarayana V S, S Ganesan, “Application of Kalman filtering technique for evaluation of neutron cross section data of ^{100}Mo ($n, 2n$) ^{99}Mo reaction”, **Nuclear Instruments and Methods of Physics Research Section A (NIM-A)**, Article No. 165850, Volume 1020, December 2021. <https://doi.org/10.1016/j.nima.2021.165850>. (Impact factor – 1.7).
2. **Sangeetha Prasanna Ram**, Jayalekshmi Nair, Suryanarayana V S, Laxman S Danu, S Ganesan, “Error Propagation using Extended Unscented Transformation Technique in Micro-correlation method for covariance analysis of efficiency of a HPGe detector”, **Nuclear Instruments and Methods of Physics Research Section A (NIM-A)**, Article No. 163057, Volume 953, February 2020. <https://doi.org/10.1016/j.nima.2019.163057>. (Impact factor – 1.455).
3. Phatak, T. S., Nair, J., **Prasanna Ram. S.**, Roy, B. J., “Evaluation of Neutron Cross-Section Data Using Gaussian Mixture Model and Digital Filter”, **Nuclear Science and Engineering**, 198(8), 1583–1606 (2023), <https://doi.org/10.1080/00295639.2023.2259748>. (Impact factor – 1.3).
4. **Sangeetha Prasanna Ram**, Jayalekshmi Nair, Vivek Singh, Dhanamma Jagli, S Ganesan, and S. V. Suryanarayana, “Application of Machine Learning algorithms for experimental data processing and estimation of ^{96}Mo (n, p) ^{96}Nb reaction cross section”, **EPJ Web of Conferences**, 284, 16005 (2023), <https://doi.org/10.1051/epjconf/202328416005>. (Impact factor – 0.334).
5. Phatak, Tejashree S., Nair, Jayalekshmi, **Ram, Sangeetha Prasanna**, Roy, B.J., and Mohanto, G., “Regression analysis of experimental reaction cross-section data of ^{241}Am ($n, 2n$) ^{240}Am ,” **EPJ Web of Conferences**, 284, 14016 (2023), <https://doi.org/10.1051/epjconf/202328414016>. (Impact factor – 0.334).
6. **Sangeetha Prasanna Ram**, Jayalekshmi Nair, Suryanarayana V S et al., “Measurement and covariance analysis of ^{100}Mo ($n, 2n$) ^{99}Mo and ^{96}Mo (n, p) ^{96}Nb reaction cross sections at the incident neutron energy of 14.54 MeV”, **Journal of Radioanalytical and Nuclear Chemistry (JRNC)**, Pages. 831-840, Volume No. 325, May 2020. <https://doi.org/10.1007/s10967-020-07213-6>. (Impact factor – 1.373).

7. **Sangeetha Prasanna Ram**, Jayalekshmi Nair, Suryanarayana V S et al., “Measurement of ^{100}Mo (n, 2n) ^{99}Mo reaction cross section and covariance analysis using extended unscented transformation technique at the incident neutron energy of 13.9 MeV”, **Indian Journal of Pure & Applied Physics (IJPAP)**, Pages 351-357, Vol. 58(05), May 2020. <http://nopr.niscair.res.in/handle/123456789/54754>. (Impact factor – 0.846).
8. Imran Pasha, Rudraswamy Basavanna, Saraswatula Venkata Suryanarayana, Haladhara Naik, **Sangeetha Prasanna Ram**, Laxman Singh Danu, Tarun Patel, Saroj Bishnoi, Manjunatha Prasad Karantha, “Measurement of neutron induced reaction cross sections of palladium isotopes at the neutron energy of 14.54 ± 0.24 MeV with covariance analysis”, **Journal of Radioanalytical and Nuclear Chemistry (JRNC)**, Pages. 175-182, Volume No. 325, May 2020. <https://doi.org/10.1007/s10967-020-07218-1>. (Impact factor – 1.373).
9. Harshavardhan Kandevekar, Sana Khan, **Sangeetha Prasanna Ram**, Jayalekshmi M. Nair, S. Ganesan "A Preliminary Examination of the Application of Unscented Transformation Technique to Error Propagation in Nonlinear Cases of Nuclear Data Science", **Nuclear Science and Engineering**, Volume 183, Pg.356–370, 2016. (Impact factor – 1.779).
10. Pradnya Jambhekar, **Sangeetha Prasanna Ram**, "EOG based Study of Eye Movements and its Application in Drowsiness Detection", **International Journal of Advanced Research in Electrical, Electronics and Instrumentation Engineering**, Vol.6, Issue 3, 2017, ISSN (Print): 2320 – 3765. (Impact factor – 5.621).
11. R. Anchan, A. Pillay, A. Kale, A. Bhadriraja and S. P. Ram, “Optimal Bipolar Lead Placement in Electrooculography (EOG): A Comparative Study with an Emphasis on Prolonged Blinks,” 2020 11th International Conference on Computing, Communication and Networking Technologies (ICCCNT), IIT-Kharagpur, India, 2020, pp. 1-7, <https://doi.org/10.1109/ICCCNT49239.2020.9225609>.

• **Patents:**

1. A NOVEL WEARABLE FOR DETECTING DROWSINESS AMONG OPERATOR OF HEAVY MACHINERY/VEHICLES, (Application No. 202121008470), 2021.
2. SYSTEM FOR PARAMETER ESTIMATION OF ELECTRONIC CIRCUIT, (Application No. 202321057357), 2023.

4. Endorsement letter from the PI and Co-PIs Institute (Please use DST/SERB format)

 **Vivekanand Education Society's**
Institute of Technology
(An Autonomous Institute Affiliated to University of Mumbai, Approved by AICTE & Recognised by Govt. of Maharashtra)

Dr. (Mrs.) J. M. Nair
M. Tech., Ph.D. (IIT-B)
Principal

Ref. No.: VESIT/JMN/PKY/1514/2025-26

Date: 18-11-2025

Endorsement from the Head of the Institution of PI

This is to certify that:

1. Institute welcomes participation of Name: **Dr. Sangeetha Prasanna Ram**, Designation: **Associate Professor** as the Principal Investigator and **Ms. Venkata Ramya Tirumalasetti** as the Co-Investigator for the project titled **"Fitonexa – The Smart Fitness Expert"** and that in unforeseen event of discontinuance by the Principal Investigator, the Co-Investigator will assume the responsibility of the fruitful completion of the project with approval of I-DAPT-HUB Foundation @ IIT (BHU) Varanasi.
2. The PI **Dr. Sangeetha Prasanna Ram** is a permanent or regular employee of this Institution and has 11 years of regular service left before superannuation.
3. The project starts from the date on which the Institution receives the grant from I-DAPT-HUB Foundation @ IIT (BHU) Varanasi.
4. The investigator will be governed by the rules and regulations of the Institution and will be under administrative control of the Institution for the duration of the project.
5. The grant-in-aid by the I-DAPT-HUB Foundation @ IIT (BHU) Varanasi will be used to meet the expenditure on the project and for the period for which the project has been sanctioned as mentioned in the sanction order.
6. No administrative or other liability will be attached to I-DAPT-HUB Foundation @ IIT (BHU) Varanasi at the end of the project.
7. The Institution will provide basic infrastructure and other required facilities to the investigator for undertaking the research project.
8. The Institution will take into its books all assets created in the above project and its disposal would be at the discretion of I-DAPT-HUB Foundation @ IIT (BHU) Varanasi.
9. The Institution assumes to undertake the financial and other management responsibilities of the project.

 **Seal of Institution**

 **Signature**

Principal,
Registrar of University/Head of the Institution
VIVEKANAND EDUCATION SOCIETY'S
INSTITUTE OF TECHNOLOGY
HASHU ADVANI MEMORIAL COMPLEX,
COLLECTOR'S COLONY, CHEMBUR,
MUMBAI-400 074, INDIA.

Date:

Hashu Advani Memorial Complex, Collector's Colony, Chembur, Mumbai, 400 074, INDIA.
Phone : +91 22 6153 2532 | Fax : +91 22 6153 2555 | Email : vesit@ves.ac.in | Website : www.ves.ac.in/vesit

5. Name of Three to Five reviewers with complete details:

1. Mr. Parag Sheth, Pacecon Technosys Pvt. Ltd.
2. Mr. Avinash Somnathan, SoC Design Engineer, Intel.
3. Mr. Rupesh Ghyar, MD, IF MedTech Pvt. Ltd. (Former Senior Executive Officer, BETIC Lab IIT-B)

5. Project Summary

Title of Project	Fitonexa – The smart fitness expert
Present TRL of the proposed work (If PI is already engaged in the proposed research)	TRL-3
Expected TRL level after completion of the project	TRL-5
The proposed work has a "Physical Cyber System" component, or fits this theme. If so, how?	The proposed work has a physical cyber system, as the proposed system collects images of the patient and gives real-time biomechanical feedback on his/her exercise regime. It also gives suggestions to improve the performance of the patient while doing physical exercise.
The proposed work has a "Data Analytics and Predictive Technologies" component, or fits this theme. If so, how?	Yes, the proposed system has Data Analytics and Predictive Technologies, as it is a portable AI-powered device providing real-time biomechanical feedback during exercise, using MediaPipe which is an open-source, cross-platform framework for building machine learning (ML) pipelines to process real-time data like video and audio.
Proposed Budget	Recurring- Rs. 2,40,000/= Non-Recurring- Rs. 5,60,000/= Overhead- Rs. 1,20,000/=
Expected Deliverable	Patents - One Publications- Two Tech. Transfer/Product Dev – N.A Manpower Training - Four

6. Assessment of Project by PI (Self-Assessment)

Parameters	Marks on 0 to 10 Scale
Suitability of the proposal for consideration under I-DAPT- HUB FOUNDATION, IIT (BHU) with respect to the identified thrust areas of the hub	9
Alignment of the proposal with the theme of Cyber Physical Systems & Data Analytics, and Predictive Technology	8
Relevance of the project for the Country's Societal / Industrial / Academic growth.	10
Contribution of the project to technology and product development.	8
Essentiality of the Equipment.	9
Competence of the investigating team (PI, Co-PI) in undertaking the proposed work.	10
Whether the proposed research has the potential for Patent/Startup/Commercialization of product/technology.	9
Scope for Industry Collaboration	9

Signature of PI and Co-PIs

Project Format: PART – II (Maximum 10 Pages)

**Call for Innovative and Entrepreneurial Project Proposals
Under
Technology Incubation Hub (TIH) of National Mission on Interdisciplinary Cyber
Physical Systems (NMICPS) for
Data Analytics and Predictive Technologies (DAPT)**

Project Ref No. _____
(to be given by the office)

1. Area Code (Please tick ✓ the appropriate box):

1.	Telecommunications	
2.	Power	
3.	Defence Research and Development	
4.	Road Transport and Highways	
5.	Health and Family Welfare	✓
6.	Any other area that fits in the National Mission on Cyber Physical Systems (NM-ICPS) - Interdisciplinary Data Analytics and Predictive Technologies (IDAPT)" theme	

2. Details of the Proposed Project:

1.	Title of the Project	Fitonexa – The smart fitness expert
2.	Present TRL of the proposed work (If PI is already engaged in the proposed research)	TRL-3
3.	Expected TRL level after completion of the project	TRL-5
4.	Project Duration	18 months
5.	Project Category (Tick ✓)	Category A ✓ / Category B / Category C

Note: Details about different categories and the duration of project proposals

Category A (TRL3 and above may apply in this category, to achieve TRL 5 and above)	18 months with a maximum budget of 15 Lakhs The project may be considered for extension with additional funds.
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	Concrete plan to achieve a minimum TRL level 5.
Category B (TRL 4 and above may apply in this category, to achieve TRL 6 and above)	Two years with a budget of up to 25 Lakhs and a concrete plan to achieve a minimum TRL level 6. Extendable to the third year based on progress. An industry partnership of 10% of the proposed budget is mandatory. One PI/Co-PI from IIT (BHU) is desirable.
Category C (TRL 5 and above may apply in this category, to achieve TRL 7 and above)	Two years with a budget of over 25 Lakhs and a concrete plan to achieve a minimum TRL level 7. Extendable to the third year based on progress. An industry partnership of 20% of the proposed budget is mandatory for up to 50 lakh projects. For a proposed budget of more than 50 Lakhs, an industry partnership of a minimum of 30% of the total proposed budget is mandatory. One PI/Co-PI from IIT (BHU) is desirable.

3. Industry Collaboration: Prosmart Concepts, (located at 208/209, Opp Deonar Bus Depot, Ujagar Chambers, Off Sion Trombay Road, Deonar, Mumbai-400088, Maharashtra, India), incorporated in Year 2011 by Mr Atul Anand (D.I E). Prosmart Concepts has agreed to collaborate and partner for the development of Fitonexa.

4. Summary of Ongoing / Completed Projects with the Principal Investigator (Last 5 Years) :
(Title, Funding Agency, Duration, Amount, Deliverables in the completed project)

S. No.	Project Title	Funding Agency	Duration	Amount in Rs.	Status and Deliverables
1	Development of Methodology for Indian Evaluated Nuclear Data Library for applications in S & T	Department of Atomic Energy, Government of India.	4 years (2019-23)	22,46,276	Completed on 31.03.2023 Research report indicating the successful development of a novel methodology for nuclear data evaluation was submitted.

5. Review of the status of Research and Development in the subject (Maximum of 1 page)

The Fitonexa project represents an innovative step in AI-driven fitness technology, integrating computer vision, machine learning and ergonomic design to create a portable, intelligent workout companion. Designed to provide real-time form correction and performance feedback, Fitonexa uses pose estimation algorithms like MediaPipe to analyze body movements with high accuracy, comparable to professional motion capture systems. Its unique combination of portability and AI-powered feedback addresses critical gaps in India's fitness technology ecosystem, offering personalized guidance, injury prevention insights and data-driven progress tracking. Some research work indicating the status of the Research and Development in this subject are listed below:

1. Computer Vision and Pose Estimation Accuracy: "Human Joint Angle Estimation Using Deep Learning and MediaPipe Framework" - Choi et al., 2024, pmc.ncbi.nlm.nih.

This study validates joint angle measurement accuracy using computer vision methods, reporting Mean Absolute Joint Angle Error (MAJAE) values between 12-44 degrees depending on the method and body movement. The research supports the project's target accuracy of $\pm 8-12^\circ$ for consumer-grade applications and demonstrates the feasibility of real-time pose estimation for exercise monitoring.

2. Real-Time Exercise Form Correction Systems: "Real-Time Posture Correction for Gym Workouts using MediaPipe Framework" - International Journal of Creative Research Thoughts, 2025.

This research demonstrates a working system using MediaPipe Pose Estimation tracking 33 anatomical landmarks at 30 FPS, achieving 95% exercise recognition accuracy and 40% reduction in form-related errors compared to unassisted training. The study directly supports Fitonexa's technical approach and expected outcomes.

3. India Fitness Technology Market Growth: "India Fitness App Market Size, Share, Growth, Trends, 2033" - IMARC Group, 2023imarcgroup

Market research shows India's fitness app market growing at 19.47% CAGR from USD 436.14 million in 2024 to USD 2,453.51 million by 2033, with AI-driven personalization being a key growth driver. This validates the commercial opportunity for Fitonexa in the Indian market.

4. Sports Biomechanics and Injury Prevention: "Cutting-Edge Research in Sports Biomechanics" - Tai et al., 2023pmc.ncbi.nlm.nih

This comprehensive review emphasizes how understanding biomechanical factors contributing to sports injuries enables development of injury prevention strategies. The research supports Fitonexa's value proposition for reducing exercise-related injuries through form correction.

5. Goniometer Accuracy and Validation Standards: "Analysis of the concurrent validity and reliability of five different clinical goniometric devices" - Kiatkulanusorn et al., 2023.

This study establishes accuracy benchmarks for joint angle measurement devices, showing excellent concurrent validity ($ICC > 0.99$) with 95% limits of agreement within $\pm 10^\circ$ for human joint measurements. The research provides validation criteria that support Fitonexa's target accuracy specifications.

6. Physiotherapy Equipment Market in India: "India Physiotherapy Equipment Market Size & Outlook, 2030" - Grand View Research, 2022.

Market analysis shows India's physiotherapy equipment market valued at USD 913.7 million in 2023, expected to reach USD 1,673.5 million by 2030 with 9% CAGR. This supports Fitonexa's target market of physiotherapy clinics and validates the B2B opportunity.

7. 3D Human Posture Detection Accuracy: "Detection of 3D Human Posture Based on Improved MediaPipe" - Scientific Research Publishing, 2023.

Research demonstrates that improved MediaPipe algorithms can achieve >90% accuracy in 3D human posture detection across people of different heights, weights, ages and genders. This supports Fitonexa's technical feasibility and adaptive calibration approach.

8. AI in Fitness and Movement Correction: "Artificial Intelligence in Fitness: Pose Estimation and Movement Correction" - Cureus Journal, 2024.

This comprehensive survey reviews AI applications in fitness, highlighting real-time feedback systems and pose correction technologies. The research validates the growing field of computer vision-based exercise monitoring and supports Fitonexa's positioning as an innovative solution.

5.1. Please explain how your proposed project fits in the "National Mission on Interdisciplinary Cyber Physical Systems (NM-ICPS) - Interdisciplinary Data Analytics and Predictive Technologies (IDAPT)" theme (Max 1 page).

Fitonexa, the proposed project, fits the ‘National Mission on Interdisciplinary Cyber Physical system – Interdisciplinary Data Analytics and Predictive Technologies (IDAPT)’ because Fitonexa is a portable AI-powered device providing real-time biomechanical feedback during exercise. Leveraging computer vision (MediaPipe Pose) on NVIDIA Jetson Nano hardware, the system detects joint angles and alerts users to form errors via visual overlays on an integrated display. The project aims to build a functional prototype demonstrating core pose estimation, angle calculation, real-time feedback capabilities for 5 common exercises.

The system employs real-time pose estimation using MediaPipe Pose on a Jetson Nano, displayed on a 10.1" LCD with a color-coded skeleton overlay (green for correct, red for incorrect) and calculates joint angles for key body parts such as the knees, hips, elbows and back. Through rule-based feedback for 5–8 exercises (including squats, push-ups, deadlifts, lunges and planks), Fitonexa enables users and physiotherapists to monitor and correct movement effectively. Compact (12"×8"×2") and battery-powered for 3–4 hours, it serves as a practical, low-cost rehabilitation and fitness tool tailored for physiotherapy clinics and similar B2B applications.

The expected outcomes of the Fitonexa project include a fully functional prototype capable of real-time operation at 10–15 FPS with $\pm 10\text{--}12^\circ$ joint angle accuracy. The system will feature an exercise database incorporating rule-based thresholds for 5–8 key movements, supported by a real-time visual feedback mechanism with latency under 100 ms. This makes the project Fitonexa a Cyber Physical System, which employs Data Analytics and predictive algorithms to real-time feedback to the patient for improving his performance. Commercially, the project aims to deliver a scalable consumer product, establishing a strong market entry strategy targeting physiotherapy clinics as the primary B2B segment.

5.2. Importance of the proposed project in the context of status (Maximum 1/2 page) (Highlight the new area or gap which will be solved in the project. This is a very important section to project the novelty of the proposal)

Addressing a Critical Healthcare Gap:

India hosts over 50,000 physiotherapy clinics, many of which depend on subjective assessment and manual goniometry for evaluating patient progress. Fitonexa addresses this gap by providing objective, quantitative joint angle measurements, enabling accurate progress tracking and visual feedback that enhances patient engagement and compliance. Its affordability (₹80K–1.2L per device) offers a cost-effective alternative to traditional manual assessments (₹30–50K/month), effectively democratizing

access to precise movement analysis. Supported by clinical evidence indicating that trainer-led programs reduce injury risks, Fitonexa delivers equivalent benefits at one-tenth the cost, revolutionizing accessibility in physiotherapy and rehabilitation.

Technical Innovation in Edge AI:

Fitonexa integrates real-time pose estimation on resource-constrained hardware (Jetson Nano, 472 GFLOPS) using TensorRT optimization to achieve efficient performance. It employs a biomechanically validated rule engine that merges exercise science with computer vision principles, while temporal smoothing algorithms mitigate sensor noise without compromising responsiveness. Furthermore, adaptive calibration customizes feedback thresholds for diverse body types, improving accuracy across users. As part of its open science initiative, Fitonexa offers an open-source framework to encourage research and innovation in exercise monitoring applications.

Commercial Viability and Scalability:

Fitonexa demonstrates clear commercial potential across multiple market segments. In physiotherapy, it replaces manual goniometers (₹5K) with an objective, automated system at an affordable one-time cost (₹50K). For home fitness users, it outperforms smartphone apps by offering offline, accurate ($\pm 8-12^\circ$) feedback on a dedicated display. Compared to premium systems like Tempo or Tonal (₹2–3L), Fitonexa provides a portable alternative at one-fourth the cost. With a manufacturing target of ₹65–85K (at 500+ unit scale) and retail pricing of ₹45–55K for consumers and ₹80–120K for clinics, it ensures 20–35% gross margin. The addressable market includes around 5,000 tech-adopting clinics and 500K premium home users, aligning with the Make in India initiative through local manufacturing, job creation and domestic AI innovation.

Academic and Research Value:

The Fitonexa project offers significant academic and experiential learning opportunities. Students and researchers gain hands-on experience in embedded AI deployment (TensorFlow Lite, TensorRT), computer vision and pose estimation algorithms, real-time system design and biomechanics-based exercise science. It also encompasses product development, user testing and market validation, and bridging technical knowledge with entrepreneurial skills. The project holds strong publication potential, with a proposed research paper titled “Real-time Pose Estimation for Exercise Monitoring on Edge Devices” aimed at ACM/IEEE conferences, contributing to global advancements in AI-driven fitness technologies

5.3 Objective (Bullet form):

Technical Objectives

- Integrate MediaPipe Pose on Jetson Nano achieving 10–15 FPS real-time inference for exercise monitoring
- Develop joint angle calculation system using 3-point vector method with temporal smoothing to reduce measurement jitter
- Create rule-based form assessment engine with biomechanically validated thresholds for 5–8 exercises
- Design real-time visual feedback UI displaying color-coded skeleton overlays, angle values and corrective text
- Implement adaptive calibration system to personalize angle thresholds based on user body proportions and flexibility

- Optimize system performance for embedded deployment: sustained 10–15 FPS, <100ms feedback latency, 3+ hour battery life

Validation Objectives

- Conduct user studies with 10+ subjects to validate accuracy (target: $\pm 10\text{--}12^\circ$ error vs manual goniometer)
- Pilot test in 3 physiotherapy clinics to assess clinical utility and gather feedback for refinement
- Demonstrate technical and commercial feasibility through working prototype and market validation data

6. Work Plan:

6.1 Methodology: (Maximum of three pages) (It should contain all the details of how each objective will be addressed. This section must be detailed and have clear plans, not vague and generalized statements. Diagrams/figures are preferred to explain the work plan.

The Fitonexa system is built around a Jetson Nano 4GB board providing 472 GFLOPS of processing power with a quad-core ARM CPU. It uses a Logitech C270 HD USB webcam capable of 720p at 30 FPS for real-time pose capture, displayed on a 7" 1024×600 HDMI touchscreen. The device is powered by a 20,000 mAh USB-C PD power bank and housed in a 3D-printed or acrylic enclosure. Connectivity and mounting are facilitated via standard HDMI and USB cables and a tripod, while storage is provided by a 64GB Class 10 SD card. To ensure stable operation, the system includes a heatsink and fan for active cooling.

Technical Approach:

The Smart Trainer system is designed as a portable, edge-based AI solution for exercise monitoring and posture correction. The primary controller is a Jetson Nano (or alternatively a high-performance Raspberry Pi 5) which interfaces with a camera to capture the user's movements in real time. The camera continuously records frames at 30–60 frames per second, which are fed to the deep learning model for pose estimation. The user initiates a workout session by selecting an exercise on the edge device interface. Once the exercise starts, the camera and optional IMU sensors begin streaming data. The IMU sensors, attached to critical joints such as knees, elbows, or waist, provide acceleration and angular velocity measurements at 100–200 Hz. This data is timestamp-synchronized with the camera frames to allow sensor fusion, improving joint angle estimation, particularly in cases of occlusion or poor lighting.

For real-time pose estimation, the system employs lightweight deep learning models such as MoveNet Lightning or BlazePose. These models are optimized for embedded devices using TensorRT for GPU acceleration. Each camera frame is preprocessed before inference: the frame is resized to the model input dimensions (typically 192×192 pixels for MoveNet), normalized to a [0,1] range and optionally background-subtracted to reduce noise. The model outputs 17 keypoints corresponding to major joints.

The Smart Trainer system requires two key computational components: First, a deep learning (DL) pose estimation model such as MoveNet or BlazePose to identify joint coordinates in the camera frames. Second, a rules-based or supervised machine learning (ML) module that processes the computed joint angles and determines whether they meet exercise-specific thresholds for proper form.

Joint Detection (DL Model)

The DL pose estimation model directly outputs the coordinates of key joints for each frame. These coordinates are used to calculate joint angles using geometric formulas.

Form Evaluation (ML/Threshold Model)

The computed joint angles are fed into an ML model or threshold-checking logic that determines if the angles are within desired ranges for the selected exercise (e.g., a knee angle in a squat should be greater than 70° and less than 90°).

This module can be a simple rules engine, or it can use a trained classifier to flag errors, give feedback and update performance scores.

Advanced ML models can also classify errors by type or severity using labeled training data.

In summary, joint coordinates are extracted by the DL model, but the logic to determine exercise correctness is handled separately, either with traditional threshold checks or a trained ML classifier that recognizes correct vs. incorrect posture from joint angles and movement patterns

Joint angle computation in a Smart Trainer system is typically performed outside the deep learning model, in post-processing on the edge device (Jetson Nano or RasPi 5), after the model outputs keypoint coordinates for detected joints. The DL model, like MoveNet or BlazePose, is primarily responsible for inferring the precise locations (2D/3D coordinates) of body keypoints in each frame the algorithm then uses these for angle calculation via vector algebra.

$$\vec{AB} = B - A, \quad \vec{BC} = C - B$$
$$\theta = \arccos \left(\frac{\vec{AB} \cdot \vec{BC}}{|\vec{AB}| |\vec{BC}|} \right)$$

where A, B, C are coordinates of the respective joints. These angles are continuously compared against predefined thresholds specific to each exercise. If an angle falls outside the desired range, a real-time correction is triggered through visual cues (LEDs or screen overlay), audio cues (buzzer or TTS voice instructions), or haptic feedback (vibration motors).

To train the deep learning model, a dataset of human poses performing the target exercises is required. For robust performance across variations in body type, clothing and lighting, a dataset of at least 5,000–10,000 annotated frames per exercise is recommended. Annotation includes 2D joint coordinates for all key points. Data augmentation techniques such as rotation, scaling, flipping and color jitter are applied to improve generalization. The dataset is split into training (70%), validation (15%) and testing (15%) sets. The model is trained using cross-entropy or mean squared error loss on joint positions, with early stopping based on validation accuracy. After training, the model is converted to TensorRT format for optimized inference on the Jetson Nano, achieving inference times below 50 ms per frame, ensuring real-time feedback.

During a workout session, each exercise produces a JSON summary file containing exercise name, user ID, number of repetitions, average joint angles, detected errors per repetition, accuracy percentage and estimated calories burned. The JSON file is stored locally on the edge device immediately after the exercise ends. If multiple exercises are performed, their summaries are appended to a single session file. This local storage approach ensures that no data is lost if Wi-Fi connectivity is unstable. At the end of the session, when the user clicks the “End Session” button, the entire session JSON is transmitted to the backend server via HTTP POST requests. The backend, implemented using Supabase or Firebase, stores the data in a structured format (e.g., PostgreSQL with JSONB columns) and allows the Flutter mobile app to retrieve and display per-exercise and session-level summaries, including tables of exercises, number of repetitions, accuracy and calories burned.

Sensor fusion between camera and IMU data is performed to enhance joint angle estimation. IMU data is first preprocessed to remove sensor bias and noise. The joint angles from the IMU are combined with

vision-based angles using a weighted average:

$$\theta_{\text{fused}} = w_v \theta_{\text{vision}} + w_i \theta_{\text{IMU}}$$

where w_v and w_i are weights tuned empirically based on validation accuracy. Optionally, a Kalman filter can be applied to smooth the fused angles and reduce jitter. This fusion ensures accurate posture detection even when joints are occluded in the camera view.

The ML model receives fused joint angles as input, which are computed from both the camera and IMU sensor data, rather than the raw data streams themselves.

Model Training and Preprocessing Details:

1) Dataset Preparation:

Curate 5,000–10,000 annotated frames per exercise, with 2D joint coordinates for all key points.

Apply data augmentation: rotation, scaling, flipping, and color jitter to improve robustness across body types, clothing and lighting.

Splitting: Divide into training (70%), validation (15%), and testing (15%) subsets.

2) Training:

i) Models are trained to minimize loss (cross-entropy for classification, mean squared error for regression on joint coordinates).

ii) Early stopping is applied if validation accuracy plateaus.

iii) Once trained, models are converted to TensorRT format for GPU-optimized inference.

3) Data Transmission:

On-device Storage during Session: As the user completes each exercise, a summary JSON file is generated locally (contains exercise name, repetitions, joint angles, errors, accuracy, calories burned).

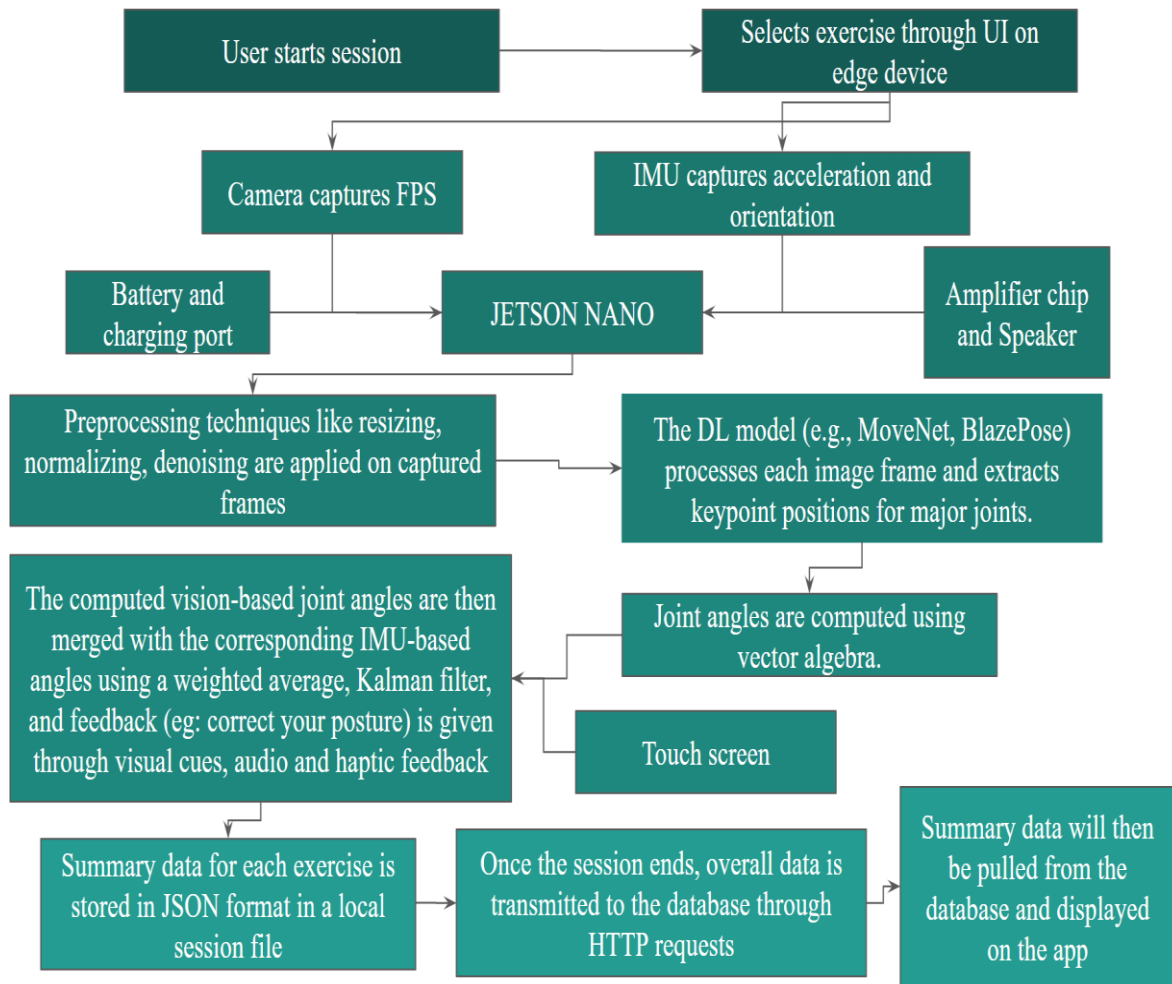
Session End: When the user ends the session, all summaries are consolidated into a single JSON session.

If a network is available, this JSON is transmitted to the backend server (Supabase or Firebase) via HTTP POST requests. If offline, the session file is queued for later transmission.

Backend Handling: Backend parses JSON, stores structured data in PostgreSQL (JSONB columns for flexibility). Flutter app fetches summaries/Tables via REST API or direct DB queries, presenting session-level stats (accuracy, reps, calories, errors).

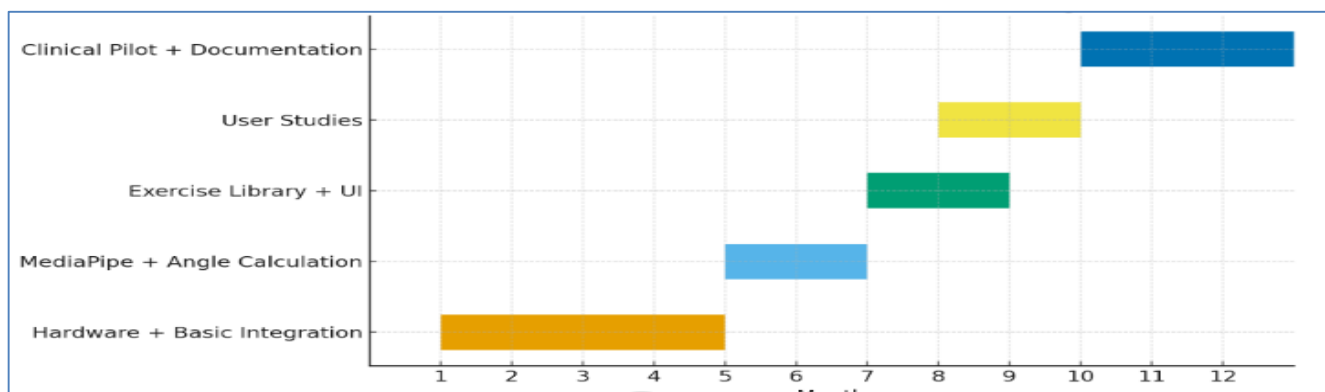
Power management is critical for a portable device. A Li-ion or LiPo battery pack with 10,000–15,000 mAh capacity, coupled with a battery management system, provides power for 2–3 hours of continuous operation. Power optimization techniques include reducing camera frame rate during idle periods, putting unused peripherals to sleep and using Jetson Nano power modes via `nvpmodel` to lower energy consumption without sacrificing inference performance.

Finally, testing and validation involve evaluating model accuracy on unseen data, measuring real-time latency from frame capture to feedback (target <200 ms), validating battery life under continuous load and performing user studies to ensure that the system correctly identifies posture errors and provides effective corrections. Once validated, the JSON summaries are displayed in the Flutter app, enabling users to track progress across exercises and sessions.



6.2 Time Schedule of activities giving milestones through the BAR diagram.

Month	Milestone	Deliverable
1–4	Hardware + Basic Integration	Assembled prototype, camera feed working
5-7	MediaPipe + Angle Calculation	Real-time pose with angles displayed
8-10	Exercise Library + UI	5 exercises, adaptive calibration
11-13	User Studies	Accuracy validation data (n=10)
14-18	Clinical Pilot + Documentation	Clinic feedback, final report, demonstration



6.3 Suggested Plan of action for utilization of research outcome expected from the project. (Maximum one page)

The research outcomes from the Fitonexa project will be strategically utilized for both academic and commercial advancement. The immediate goal is to refine the functional prototype based on experimental validation and user feedback from physiotherapy clinics. Data collected during testing will be analyzed to improve model accuracy, robustness and feedback algorithms. Once validated, the system design, algorithms and calibration methods will be documented and published in open-access repositories to enable reproducibility and support further research in real-time human motion analysis.

In parallel, collaboration will be initiated with physiotherapy centers, rehabilitation startups and fitness equipment manufacturers to pilot the system in real-world use cases. A technology transfer or incubation plan will be developed through institutional innovation cells or startup incubation centers to support commercial deployment and product certification (e.g., CE/FDA if applicable). Additionally, the research outcomes will contribute to academic learning through workshops, student projects and publications in IEEE/ACM conferences focused on edge AI and biomechanics.

The long-term plan involves scaling production via local manufacturing partnerships under the Make in India initiative, enabling large-scale deployment to physiotherapy clinics, sports centers and home fitness users. The dataset and framework developed will also serve as a foundation for advanced research in personalized exercise analytics, rehabilitation monitoring and AI-assisted healthcare devices.

7. Expected Deliverables:

Technologies, Products, IPRs:	Technology; 1. Real-Time Pose Estimation System Optimized for Edge Devices 2. Adaptive Biomechanical Calibration Framework Product: 1. Fitonexa Portable Exercise Monitor – Hardware 2. Software & Firmware Package
Research articles:	Two Research Paper publications
Entrepreneurship Development/ techno-commercial incubation:	Yes (Commercialization of Fitonexa software - downloadable paid App)
Startups and Spin-off companies:	Yes (Startup for design and development of Fitonexa hardware)
Training / Employment generation:	Yes

8. Risk analysis: (Maximum ½ page)

The technical risks associated with the project primarily concern system performance, accuracy, and hardware reliability. A moderate risk is maintaining a sufficient frame rate, as FPS below 10 can impact real-time analysis; this will be mitigated through resolution reduction and TensorRT optimization to ensure smooth operation. Although low in probability, insufficient accuracy beyond $\pm 15^\circ$ could critically affect motion tracking precision, so validation with a goniometer and temporal smoothing will be implemented. To prevent medium-level risks such as Jetson Nano overheating, active cooling and temperature monitoring are planned. Additionally, the possibility of reduced battery life (<2 hours) will be mitigated by using a 20,000 mAh power bank, while camera occlusion—considered a high-probability, medium-impact issue—will be managed by guiding users on optimal positioning and skipping frames when visibility drops below 60%.

The project execution risks focus on schedule adherence, resource availability, and team management. Moderate delays in the project timeline could occur due to dependencies between tasks; this risk will be reduced through parallel work streams and a built-in two-week buffer. If clinic access for data collection is denied, an alternative plan to use a college gym or fitness club will ensure continuity. While team member dropout has a low probability, its high potential impact on progress necessitates preventive measures such as clearly defined roles, cross-training among members, and maintaining comprehensive documentation to ensure smooth handover and sustained project momentum.

9. Budget:

Head	First Year	Second Year	Third Year	Total Grant (in INR only)	Justification
Recurring (Max 25% of the total budget) (Consumables, Contingencies, Travel, etc.)	1,20,000	1,20,000	--	2,40,000	Consumables for user study and validation
Non-Recurring (Equipment)	4,60,000	1,00,000	--	5,60,000	Depth Camera, NVIDIA Jetson Nano board, prototype development
Overhead (Max 10%)	60,000	60,000	--	1,20,000	
Total	6,40,000	2,80,000	--	9,20,000	
Manpower (Budget will be provided separately in the form of CHANKYA Fellowship)	Apply for the Chanakya Fellowship / Chanakya Top fellowship linking with this project. Please go through the guidelines available on the I-DAPT website for details.				
Short Term Course (STC)	STC format is available on the I-DAPT website. PI can apply as per the STC guidelines. Approval of STC will be provided along with the project, but fund transfer will be according to STC guidelines.				

10. Expertise:

10.1 Expertise available with the investigators in executing the project: (Maximum 1 page)
(Professional expertise existing with each of the investigators in terms of publications (provide only number), Patents (provide only number) and preliminary results, to execute every component of the proposal should be highlighted)

The Principal Investigator, **Dr. Sangeetha Prasanna Ram**, holds a Ph.D. in the application of **Machine Learning and Kalman Filtering techniques to nuclear data analysis**. Over the past five years, she has actively guided both undergraduate and postgraduate students on projects involving Machine Learning and Deep Learning applications in healthcare data analytics. As a preliminary study, the PI and Co-PI have successfully mentored an undergraduate project titled ‘Exercise evaluation using smart trainer system’ during the year 2024-25, which employed a software algorithm that was able to classify three exercise postures with an accuracy of approximately 90% based on joint angle analysis and position tracking. The PI’s sustained research efforts have led to the publication of several high-quality papers in international, peer-reviewed journals indexed in Scopus, Web of Science, and other reputed databases, demonstrating her consistent contributions to emerging interdisciplinary research areas. She has also published a patent on “A NOVEL WEARABLE FOR DETECTING DROWSINESS AMONG OPERATOR OF HEAVY MACHINERY/VEHICLES” (Application No. 202121008470) in the year 2021 and filed for a patent on “SYSTEM FOR PARAMETER ESTIMATION OF ELECTRONIC CIRCUIT” (Application No. 202321057357) during the year 2023.

11.1 Infrastructural facilities available with PI:

The institute is equipped with robust infrastructural facilities that strongly support advanced research and innovation. It houses well-established laboratories with state-of-the-art computing resources, modern instrumentation, and specialized software essential for data-driven research. The institute also provides high-speed internet connectivity, secure data storage systems, and access to licensed simulation and analytics tools. In addition, a well-stocked central library with extensive digital resources, journal subscriptions, and online databases ensures comprehensive scholarly support. These facilities collectively create a conducive environment for executing high-impact research projects and achieving the proposed research objectives efficiently.

11.2 Equipment available with the Institute/ Group/ Department/ for the project: (Related to this project)

The institute is equipped with the following equipment required for the project:

1. NVIDIA Jetson Nano 4GB board with quad-core ARM CPU.
2. Logitech C270 HD USB webcam
3. PC with MediaPipe Pose
4. 10.1” LCD display

PART –III (If applicable) – CHANAKYA Fellowship / Short Term Course

PI can share the details of the students he wants to engage/propose for the CHANAKYA top-up fellowship. The top-up fellowship is primarily for the IIT (BHU) students. However, the I-DAPT can consider requests for Top-up fellowships for students of other institutions if a high TRL (Above 5), technology development, transfer, or potential startup is visible/expected from the project. The students can avail the Chanakya top-up fellowship along with a regular fellowship of their respective institutes for doing work related to the project. Please see the available **CHANAKYA top-up fellowships** in the CHANAKYA fellowship section on the I-DAPT website.

For the CHANAKYA Regular Fellowship, the PI has to select a suitable candidate as per I-DAPT guidelines and their institute norm (if manpower in the form of CHANAKYA fellowship is approved for the project by I-DAPT). Final approval of I-DAPT is required for selected fellows, and then funds for the fellowship will be released.

Three UG students of third year Automation and Robotics engineering of the institute, have shown interest in working on the project. The PI would like to propose their names for the CHANAKYA Regular Fellowship for the period of the project work.

PIs can plan a **Short Term Course (STC)** in the area of the project as per the guidelines and the format available on the I-DAPT website. The project with the STCs component will be given weightage in the funding. Approval of STC will be provided along with the project, but fund transfer will be according to STC guidelines.

Note: Please go through the guidelines carefully.